Geometry Mediates Excitation, Resonance, and Identifiability in Fluid-Filled Viscoelastic Shells

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Abstract

The resonant behaviour of fluid-filled viscoelastic shells arises repeatedly in biomechanics, postharvest sensing, and non-destructive evaluation, yet each application is typically treated in isolation. This paper advances a unifying thesis: *shell geometry simultaneously filters the forcing field, organises the modal spectrum, and determines the invertibility of the corresponding parameter-estimation problem.*

We formalise this thesis through four results. First, we prove that equivalent-radius (spherical) reductions of non-spherical shells are *rank-deficient* in the sense of structural identifiability: geometry and material parameters become confounded, rendering independent recovery impossible. Second, we show that asphericity lifts this rank deficiency through curvature-dependent mode splitting, providing the additional spectral information needed for unique inversion. Third, we derive the near-spherical conditioning asymptotics that govern the transition from identifiable to non-identifiable regimes, classifying shells by their curvature parameter ε. Fourth, we demonstrate that forward-model adequacy does not imply inverse-model adequacy: an equivalent-sphere model may predict resonant frequencies within engineering tolerance yet fail catastrophically when used for parameter estimation.

These four results are supported by cross-application evidence drawn from the human abdomen (infrasound resonance), the watermelon (acoustic ripeness assessment), and the urinary bladder (fill-dependent resonance), demonstrating that geometry is the hidden variable unifying apparently disparate vibroacoustic phenomena.

Keywords: fluid-filled shells; viscoelastic resonance; structural identifiability; inverse problems; asphericity; Kac problem

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1 Introduction

Spectral questions recur across mathematics, physics, and engineering. Given an observed set of resonances, what can one infer about the system that produced them? Kac's celebrated formulation — "Can one hear the shape of a drum?" — remains the canonical version of this problem [7]. Its enduring influence lies not in the narrow case of planar drums, but in the larger programme it inaugurated: to understand which features of a physical system are encoded in spectral data, which are hidden by symmetry, and which are unrecoverable without additional information. The classical counterexample of isospectral but non-congruent domains [5] did not end that programme. It sharpened it. Inverse vibration theory, model updating, and elastographic parameter recovery all ask variations of the same question: when does a resonance spectrum carry enough independent information to support reliable inference about geometry, material properties, or both [3, 4, 20, 2, 1]?

In this paper we ask that question for fluid-filled viscoelastic shells. This class of systems appears in apparently disparate settings: the human abdomen and bladder in biomechanics, gas-pocket and sub-bass coupling in physiological acoustics, and fruit-ripeness assessment in postharvest sensing [12, 16, 15, 19, 13]. Yet these applications are linked by one structural fact. The shell geometry does not merely set a boundary condition for a forward calculation. It mediates the entire problem. Geometry filters how external forcing couples into admissible deformation fields; it organises the modal spectrum through curvature, symmetry, and fluid loading; and it determines whether the resulting spectrum can be inverted to recover the parameters that generated it. Our central thesis is therefore simple: *in fluid-filled viscoelastic shells, geometry does three jobs — it filters forcing, organises spectra, and determines invertibility.*

The first of these roles emerges on the excitation side. Paper 1 showed that a fluid-filled oblate shell can possess a genuine low-frequency flexural resonance near 4 Hz, yet airborne infrasound couples to that mode vanishingly weakly because the long-wavelength forcing field is itself filtered by geometry [12]. The same shell, driven mechanically through whole-body vibration, responds far more strongly because that pathway is not subject to the same spatial-harmonic suppression. Paper 6 extended the same logic below 20 Hz, where structure-borne pathways again dominate airborne ones [19]. The point is broader than the brown-note problem that motivated the programme. A resonance is not an effect in isolation. It matters how forcing is projected onto the mode shapes permitted by the geometry.

The second role is spectral organisation. Fluid-filled shells do not possess a featureless ladder of frequencies. Curvature, aspect ratio, internal loading, and wall compliance sort the admissible motions into qualitatively distinct families and shift their ordering in application-specific ways [9, 24, 6]. In the abdominal model, this distinguishes low-frequency flexural modes from the much higher breathing mode dominated by the compressibility of the entrained fluid [12]. In the bladder, changing fill volume alters both size and effective stiffness, producing a non-trivial resonance trajectory rather than simple monotonic scaling [15]. In borborygmi, constrained gas pockets introduce a different but related spectral mechanism, in which shell confinement selects the audible range of bubble oscillations [11]. In watermelon tap-tone analysis, the measurable spectral shifts used informally by growers are again organised by geometry and shell mechanics rather than by material changes alone [13]. Across domains, geometry is the quiet organiser of what the spectrum is allowed to look like.

The third role is inverse. A forward model may predict frequencies accurately enough for

classification, explanation, or design, yet still fail as an inversion model because the spectrum does not separate the parameters cleanly. That distinction emerged most clearly in the fruit and organ inverse problems of Papers 7 and 8. Equivalent-radius reductions can reproduce leading-order resonance trends remarkably well, which explains their practical appeal. But by collapsing shape information into a single scalar radius they confound geometric and material effects, and the inverse map becomes nearly singular or formally rank deficient [13, 14]. Recovering invertibility requires more than a good forward fit. It requires modal data in which different modes sample curvature differently. That requirement is geometric before it is statistical.

The capstone develops this programme around four conjectures, each of which captures one aspect of the geometry–spectrum–inversion relation. The first is a *rank-deficiency conjecture*: equivalent-radius models of non-spherical fluid-filled shells generically collapse independent geometric information and therefore produce ill-posed spectral inversion problems. The second is an *identifiability-lifting conjecture*: asphericity restores invertibility because distinct modes respond differently to distinct curvature channels. The third is a *near-spherical conditioning conjecture*: the transition from recoverable to unrecoverable inversion is controlled by regular asymptotics near the spherical limit, where symmetry suppresses the very sensitivities on which inference depends. The fourth is a *forward–inverse adequacy conjecture*: a model may be adequate for predicting resonant frequencies while remaining inadequate for parameter recovery. Taken together, these conjectures recast several apparently separate Browntone results as instances of one general statement about how geometry governs both excitation and inference.

The evidence for that statement was assembled in three acts. Act I, Papers 1–2, posed the question. The brown-note analysis established the coupling disparity between airborne and mechanical excitation, while the gas pocket study showed that alternative resonant substructures can couple far more efficiently than the whole cavity [12, 16]. Act II, Papers 3–6, broadened the evidence base: scaling laws showed that the dimensionless resonance structure persists across body sizes; bladder resonance revealed geometry–stiffening competition in another organ; borborygmi provided an audible constrained-bubble counterpart; and sub-bass coupling extended the excitation analysis to lower frequencies and alternative pathways [18, 15, 11, 19]. Act III, Papers 7–8, supplied the theoretical turn. Watermelon acoustics demonstrated that shell-resonance inversion can succeed when the geometry leaves enough spectral structure to exploit, while the Kac paper showed that asphericity can lift inverse degeneracy and that near-spherical models can be forward-useful yet inverse-fragile [13, 14]. The present paper treats those results not as a sequence of curiosities, but as evidence for a single geometry-mediated theory of excitation, resonance, and identifiability.

Our aim is therefore synthetic rather than merely archival. We do not repeat the detailed derivations of the companion papers. Instead, we extract the shared mathematical structure, state the four conjectures in a common language, and show how the examples together support a general picture of fluid-filled shells as systems in which symmetry and curvature govern both what can be excited and what can be inferred. The intended scope is wider than the original applications. The same logic should matter wherever one seeks to infer geometry or material properties from resonance data in soft shell-like structures, whether in biomechanics, non-destructive evaluation, or allied inverse problems.

The paper is organised as follows. Section 2 summarises the programme-level evidence from Papers 1–8 and identifies the common geometric structure that links them. Section 3 formulates the four conjectures precisely and relates them to the underlying shell and inverse problem theory.

Section 4 gathers the principal cross-paper results into a unified comparative account, emphasising where excitation, spectral organisation, and identifiability align or diverge. Section 5 considers the broader implications for spectral inference in fluid-filled shells and for the use of reduced models in inverse settings. Section 6 closes with the capstone claim: geometry is not a secondary detail of these systems, but the mechanism that mediates their forcing, their resonances, and the possibility of hearing the parameters that made them.

2 Background

2.1 Shell-model overview

The companion papers share a common forward model: a fluid-filled viscoelastic shell whose undeformed midsurface is represented, to leading order, as an oblate spheroid with semi-major axis a and semi-minor axis c. The modelling strategy follows the classical shell-acoustics tradition from Lamb's treatment of elastic-shell modal families [8], through Love–Kirchhoff shell theory [10], to later fluid–structure formulations in which structural inertia, fluid loading, and curvature appear on equal footing [6]. Paper 1 adapted that framework to the human abdomen and used a Rayleigh–Ritz approximation to retain the features that matter for the present programme: curvature, wall compliance, internal loading, and mode shape [12]. The capstone does not repeat that derivation. It uses the resulting structure as the common language in which the subsequent applications can be compared.

In schematic form, if $\mathbf{q}$ denotes the vector of Ritz amplitudes, the kinetic and potential energies are written as

$$T = \frac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}} \left(\mathbf{M}_s + \mathbf{M}_f\right) \dot{\mathbf{q}}, \qquad V = \frac{1}{2}\mathbf{q}^{\mathsf{T}}\mathbf{K}\mathbf{q}, \tag{1}$$

where $\mathbf{M}_s$ is the shell inertia matrix, $\mathbf{M}_f$ is the added-mass contribution from the entrained fluid, and $\mathbf{K}$ is the effective stiffness matrix arising from membrane strain, bending, and any prestress retained in the model. Stationarity of the Rayleigh quotient then leads to the generalised eigenvalue problem

$$\left[\mathbf{K} - \omega^2 \left(\mathbf{M}_s + \mathbf{M}_f\right)\right] \mathbf{q} = 0, \tag{2}$$

where ω is the angular natural frequency. For nearly spherical shells this structure reduces to the classical modal families identified by Lamb [8]; for the more general fluid-loaded case, it fits squarely within the coupled shell–acoustic framework described by Junger and Feit [6]. The important point for the present paper is that geometry enters twice. It shapes the admissible deformation fields through the basis itself, and it enters the energetic balance through curvature-dependent stiffness and fluid coupling. The same geometric ingredients later control both forcing efficiency and inverse conditioning.

2.2 Companion papers: Act I — the question

Act I asked whether the notorious brown-note claim contained any physically respectable core, and the answer turned out to be more interesting than the folklore. Paper 1 showed that the abdomen, modelled as a fluid-filled oblate shell, does possess a genuine low-frequency flexural

resonance, with the second mode occurring at

$$f_2 = 3.95\,\mathrm{Hz}. \tag{3}$$

That result mattered because it separated two questions that are often conflated: whether a resonance exists, and whether any realistic forcing path can excite it appreciably. The modal calculation answered the first question in the affirmative. The coupling analysis answered the second in the negative for airborne infrasound. Because the acoustic field is both long-wavelength and weakly matched to the relevant flexural deformation, the airborne pathway does little work on the shell. Mechanical forcing through whole-body vibration projects onto the same mode family far more effectively, producing a coupling ratio of approximately $R \approx 3.3 \times 10^4$ between the mechanical and airborne responses [12]. The resonance was real; the direct acoustic route to it was not.

Paper 2 supplied the first important refinement [16]. Instead of treating the abdominal cavity as the only relevant resonant structure, it considered tissue-constrained gas pockets as local compliant inclusions capable of much stronger pressure-to-motion transduction. This did not contradict the whole-cavity result of Paper 1. It clarified it. If one seeks a global flexural response of the abdominal shell, airborne excitation is ineffective. If one asks whether smaller substructures embedded within the abdomen might respond locally, gas pockets are more plausible candidates. The programme therefore began with a division that remains useful throughout the capstone: global shell resonance organises the large-scale spectrum, but local resonant substructures may dominate transduction when coupling efficiency is the relevant physiological question.

2.3 Act II — evidence across scales, organs, and pathways

Act II tested whether the same shell-theoretic logic survived changes in scale, organ, and excitation pathway. Paper 3 recast the abdominal resonance problem in dimensionless form and derived scaling laws that made the low-order flexural behaviour comparable across species [18]. Its point was not merely zoological. By showing that the governing frequency scales can be expressed through a compact set of geometric and mechanical groups, the paper demonstrated that the infrasonic shell problem is structured by similarity, rather than by the contingencies of one anatomical example.

Paper 4 transferred the framework to the urinary bladder and thereby tested it on an organ whose geometry changes substantially with physiological state [15]. The bladder calculations showed that fill-dependent resonance cannot be interpreted as a simple size effect. Increasing volume changes characteristic length scales, but it also alters wall tension and effective stiffness. The resulting modal trajectory therefore reflects a competition between geometry and stiffening, not a monotonic geometric trend. This was the first clear sign, outside the original abdominal problem, that a single scalar reduction of shape is unlikely to preserve the physically relevant sensitivities.

Paper 5 shifted attention from infrasonic flexure to an audible internal phenomenon: borborygmi [11]. Its constrained-bubble model produced characteristic frequencies in the range 135 Hz to 440 Hz, matching the clinical band of gut sounds. The mechanism was distinct from whole-cavity flexure, but the shell still mattered. It set the confinement and loading conditions within which local gas pockets oscillate. The paper therefore broadened the programme without

abandoning its central theme: geometry continues to mediate what kind of spectrum is possible, even when the relevant source is local rather than global.

Paper 6 returned to low frequencies and extended the coupling question to the sub-bass regime, where concert exposures and whole-body vibration are often discussed together despite acting through rather different pathways [19]. The conclusion was consistent with Paper 1. Airborne sub-bass remains a poor driver of substantial abdominal motion, whereas structure-borne excitation can couple much more efficiently. By the end of Act II, the programme had assembled a coherent evidential base: across species, organs, audible and infrasonic ranges, and airborne and mechanical forcing, geometry continued to organise the modal spectrum while filtering how energy enters it.

2.4 Act III — from forward adequacy to inverse structure

Act III turned from forward resonance prediction to the inverse question of what one can infer from a measured spectrum. Paper 7, on watermelon tap-tone assessment, provided the constructive example [13]. It showed that an equivalent-sphere shell model can be forward-adequate for a practical inversion task when the inferential target is narrow and the geometric uncertainty is limited. In that context the model need only map a measured resonance to an effective rind stiffness with useful fidelity. The watermelon paper therefore demonstrated that reduced geometry can be serviceable when the application asks for classification or approximate property recovery, rather than full geometric disentanglement.

Paper 8 exposed the inverse price of that simplification [14]. Asked in Kac-like language whether one can "hear" the shape of an organ, the programme's answer became conditional rather than categorical. If the shell is reduced to a single equivalent radius, material and geometric sensitivities become strongly correlated, and the inverse map is nearly singular. The condition-number analysis showed that the equivalent-sphere model is catastrophically ill-conditioned, whereas the full oblate-shell model restores practical separation of parameter effects. Just as importantly, Paper 8 revealed an oblate–prolate asymmetry: breaking spherical symmetry is not, by itself, enough. What matters is whether the perturbation changes how different modes sample curvature.

Paper 9 condensed that observation into a short theorem-oriented note on identifiability lifting [17]. Its contribution was to show that oblate deformations lift the near-spherical inverse degeneracy in a way that prolate deformations do not improve comparably. This matters to the capstone because it rules out an easy but misleading slogan. One cannot simply say that "asphericity helps". Some geometric departures from the sphere decorrelate the spectral sensitivities that matter for inversion; others do not. The programme's theoretical turn therefore culminated in a sharper claim: forward adequacy and inverse adequacy are distinct, and their separation is governed by geometry.

2.5 Broader context

The Browntone series sits at the meeting point of several established literatures. One is classical shell acoustics, in which curvature, modal families, and fluid loading have long been treated as inseparable elements of the forward problem [10, 9, 24, 6]. Another is spectral inverse theory, from Kac's celebrated question and the Gordon–Webb–Wolpert counterexample to engineering treatments of vibration inversion and model updating [7, 5, 3, 4, 20, 2]. A third is the practical

language of identifiability, familiar from inverse elasticity and biomedical imaging, in which the issue is not merely uniqueness in principle but whether the measured observables contain enough independent information to distinguish the parameters of interest under realistic sensitivity and noise structures [22, 1, 21, 23].

What the present programme adds is a concrete family of case studies in which these literatures can be read together. The same geometric framework explains why a soft organ can possess a low-frequency shell resonance yet remain almost immune to direct airborne excitation; why local gas pockets and gut sounds are better interpreted as shell-constrained substructures than as whole-cavity breathing modes; why a reduced model can succeed as a forward surrogate for fruit-ripeness assessment; and why that same reduction can fail badly when used for multi-parameter inversion. In biomedical and engineering settings alike, the practical question is rarely whether resonance exists in the abstract. It is whether the observable spectrum contains geometry-specific information that survives modelling reduction and experimental uncertainty. The companion papers suggest that this question is best answered not by separating shell mechanics, acoustics, and inversion into different problems, but by treating geometry as the common mediator of all three.

3 Theory

The companion papers established three ingredients that we now place in a common mathematical frame. Paper 1 distinguished the low-frequency flexural family from the much higher breathing mode in a fluid-filled soft shell, and showed that geometry controls not only resonance but also the efficiency with which forcing projects onto those modes [12]. Paper 7 showed that an equivalent-sphere model can nevertheless be operationally useful when the inverse task is deliberately narrow and the geometry is known independently [13]. Paper 8 then exposed the inverse price of that reduction: once geometry and material must be inferred simultaneously, the same equivalent-radius simplification becomes structurally defective [14]. The present section formalises that programme as four conjectures.

We emphasise at the outset that the statements below concern *local practical identifiability* of flexural spectra. They do not claim global uniqueness for arbitrary nonlinear inversions. They ask a more disciplined question: given a nominal operating point, do the measured modal frequencies contain enough linearly independent information to distinguish changes in geometry from changes in material stiffness?

3.1 Common setup

Consider a fluid-filled viscoelastic shell with semi-major axis a, semi-minor axis c, Young's modulus E, and fixed secondary parameters

$$\mathbf{p} = (h, \nu, \rho_w, \rho_f, K_f, P_{\mathrm{iap}}, \eta).$$

For the Browntone applications, the canonical values are $E = 0.1\,\mathrm{MPa}$, $a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 0.01\,\mathrm{m}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$, $K_f = 2.2\,\mathrm{GPa}$, $P_{\mathrm{iap}} = 1000\,\mathrm{Pa}$, and $\eta = 0.25$ [12, 14]. We restrict attention to the flexural modes $n \geq 2$, since the breathing mode $n = 0$ is fluid-compressibility dominated and lies in the kilohertz range for the canonical abdomen

rather than the single-digit-hertz range of interest [12]. That distinction is not decorative; it keeps the physics honest.

Let the unknown parameter vector be

$$\boldsymbol{\theta} = (a, c, E)^\top \in \Theta \subset \mathbb{R}^3_+, \tag{4}$$

and let the measured modal set be $\mathcal{M} = \{n_1, \ldots, n_m\}$, with $m \geq 3$. The forward map is

$$\mathbf{f}(\boldsymbol{\theta}) = \big(f_{n_1}(\boldsymbol{\theta}; \mathbf{p}), \ldots, f_{n_m}(\boldsymbol{\theta}; \mathbf{p})\big)^\top \in \mathbb{R}^m, \tag{5}$$

where f_n is the natural frequency of flexural mode n. The raw Jacobian is

$$J_{ij} = \frac{\partial f_{n_i}}{\partial \theta_j}, \tag{6}$$

but because the three parameter directions carry different units, the relevant local sensitivity operator is the *scaled Jacobian*

$$J_s[i, j] = \frac{\partial f_{n_i}}{\partial \theta_j} \frac{\theta_j}{f_{n_i}}. \tag{7}$$

Each entry of J_s is dimensionless and measures fractional frequency change per fractional parameter change. Let

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0 \tag{8}$$

denote the singular values of J_s, and define the local conditioning metric

$$\kappa(J_s) = \frac{\sigma_1}{\sigma_3}. \tag{9}$$

When $\sigma_3 = 0$, the inverse problem is locally rank deficient. When $\sigma_3 > 0$ but small, the inverse is formally possible but practically fragile: small frequency errors may be amplified into large parameter errors. Thus the smallest singular value identifies the least observable parameter combination, while κ measures how badly the geometry conspires against the experimentalist.

3.2 Equivalent-radius reductions and lost geometric dimension

Equivalent-sphere models replace the two-parameter geometry (a, c) by a single scalar descriptor, typically the equivalent radius

$$R_{\text{eq}} = (a^2 c)^{1/3}, \tag{10}$$

or, more generally, some smooth scalar reduction $g(a, c)$. Such reductions can be entirely sensible for forward estimates of resonance magnitude or scaling, and indeed they are useful in Papers 1 and 7 for precisely that purpose [12, 13]. The inverse consequence is less cheerful.

Proposition 1 (Rank collapse under scalar geometric reduction) *Let a reduced forward model satisfy*

$$f_n(a, c, E; \mathbf{p}) = \phi_n\big(g(a, c), E; \mathbf{p}\big) \qquad \textit{for all } n \in \mathcal{M}, \tag{11}$$

with $g : \mathbb{R}_+^2 \to \mathbb{R}_+$ a smooth scalar function. Then the two geometric columns of the Jacobian are everywhere collinear, so that

$$\operatorname{rank}\left[\frac{\partial \mathbf{f}}{\partial a}, \frac{\partial \mathbf{f}}{\partial c}\right] \leq 1, \tag{12}$$

and consequently

$$\operatorname{rank}(J) \leq 2, \qquad \operatorname{rank}(J_s) \leq 2. \tag{13}$$

Justification. By the chain rule,

$$\frac{\partial f_n}{\partial a} = \frac{\partial \phi_n}{\partial g}\frac{\partial g}{\partial a}, \qquad \frac{\partial f_n}{\partial c} = \frac{\partial \phi_n}{\partial g}\frac{\partial g}{\partial c}.$$

Hence both geometric columns are scalar multiples of the same mode vector $\left(\partial \phi_{n_i}/\partial g\right)_{i=1}^m$. For the specific choice $g = R_{\mathrm{eq}} = (a^2 c)^{1/3}$, one obtains

$$\frac{\partial R_{\mathrm{eq}}}{\partial a} = \frac{2}{3}\frac{R_{\mathrm{eq}}}{a}, \qquad \frac{\partial R_{\mathrm{eq}}}{\partial c} = \frac{1}{3}\frac{R_{\mathrm{eq}}}{c}, \tag{14}$$

and therefore the scaled columns satisfy $J_s[:, a] = 2\, J_s[:, c]$ exactly. One geometric degree of freedom has been removed before any data are fitted.

Conjecture 2 (Rank-deficiency of equivalent-radius models) *For any fluid-filled shell model in which the geometry enters the flexural spectrum only through a scalar reduction $g(a, c)$, the local inverse problem for $\boldsymbol{\theta} = (a, c, E)$ is structurally rank deficient. In particular, the scaled Jacobian with respect to (a, c, E) satisfies*

$$\operatorname{rank}(J_s) \leq 2, \tag{15}$$

so that one independent combination of (a, c) is unobservable from the modal frequencies alone. Equivalent-radius reductions therefore cannot support independent recovery of both geometric axes and material stiffness, regardless of how accurately the frequencies are measured.

The conjecture is stronger than a complaint about poor conditioning. It states that the missing information is *structural*: no amount of experimental precision can recover a geometric direction that the model has collapsed into a single scalar. One may still estimate $g(a, c)$ and E, but not a and c separately. Paper 8 exhibited this numerically for the equivalent-sphere reduction, where the nominal condition number reached $\kappa \approx 1.37 \times 10^{10}$, finite only because the derivatives were approximated numerically rather than taken symbolically [14].

3.3 Asphericity as an identifiability-lifting perturbation

To escape Conjecture 2, the model must retain at least two genuinely independent geometric channels. For an oblate shell this occurs through curvature anisotropy. Writing the eccentricity as

$$\varepsilon = \sqrt{1 - c^2/a^2}, \tag{16}$$

the spherical limit is $\varepsilon = 0$, while $\varepsilon > 0$ introduces distinct meridional and circumferential curvature distributions. In a Ritz or perturbation formulation, mode n samples those curvature fields through mode-specific weighting functions. Schematically,

$$f_n(\varepsilon) = f_n^{(0)} + \alpha_n \varepsilon^2 + \beta_n \varepsilon^4 + \cdots, \tag{17}$$

where the coefficients $\alpha_n, \beta_n, \ldots$ depend on the mode number because different mode shapes concentrate strain energy in different regions of the shell. The key point is not the polynomial form by itself, but the variation of these coefficients with n. Once that happens, the a- and c-columns of the Jacobian need no longer remain proportional across modes.

Remark 3 *The relevant mechanism is* mode-dependent curvature sampling. *Low-order and higher-order flexural modes do not interrogate the same parts of an oblate shell with the same weights, so the spectrum acquires more than one geometric signature. Geometry ceases to be a single number and resumes its full-time occupation.*

Conjecture 4 (Identifiability lifting by asphericity) *For an oblate family of fluid-filled viscoelastic shells with $\varepsilon > 0$, retention of the true spheroidal curvature in the forward model lifts the equivalent-radius degeneracy. More precisely, for a generic set of measured flexural modes $\mathcal{M}$ containing at least three modes, there exists an open set of oblate operating points on which*

$$\operatorname{rank}(J_s) = 3, \tag{18}$$

because different modes weight the meridional curvature field differently and thus produce non-proportional geometric sensitivity columns. At the canonical abdominal operating point studied in Paper 8, this lifting is substantial: the condition number falls from the equivalent-sphere value $\kappa \approx 1.37 \times 10^{10}$ to the oblate Ritz value $\kappa = 69.4$ [14].

This conjecture does not say that every aspherical shell is automatically well-conditioned. It says that oblate asphericity restores the *possibility* of independent recovery by breaking the exact proportionality imposed by scalar geometric reduction. Conditioning then becomes a quantitative matter rather than an algebraic impossibility.

3.4 Near-spherical asymptotics and the absence of a true power law

The next question is how the restored identifiability behaves as the shell approaches spherical symmetry. The natural temptation is to seek a singular power law $\kappa \sim \varepsilon^{-\alpha}$ as $\varepsilon \to 0$. Paper 8 showed that this intuition is misleading for the oblate Ritz family [14]. Because the oblate perturbation is reflection-symmetric, odd powers of ε are suppressed, and the smallest singular value is better represented by a regular even expansion than by a singular asymptotic blow-up.

Conjecture 5 (Near-spherical conditioning asymptotics) *For the oblate spheroidal shell family with a curvature-resolving forward model, the smallest singular value of the scaled Jacobian admits a regular expansion of the form*

$$\sigma_3(\varepsilon) = \sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4), \qquad \varepsilon \to 0, \tag{19}$$

with $\sigma_0 > 0$. Consequently,

$$\kappa(J_s) = \frac{\sigma_1(\varepsilon)}{\sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4)} \tag{20}$$

has a finite near-spherical floor rather than a true divergence, so there is no genuine asymptotic power law in the oblate class. In the canonical computations of Paper 8, the directly evaluated near-spherical

floor was $\kappa \approx 269$ at $\varepsilon = 0.01$, while an intermediate-range fit over $\varepsilon \in [0.15, 0.85]$ gave $\sigma_0 = 0.01113$ and $\lambda_1 = 0.00507$, so that the crossover $\varepsilon_c = \sqrt{\sigma_0/\lambda_1} = 1.48$ lies outside the physical oblate range [14]. By contrast, the prolate family does not exhibit the same oblate lifting trend: its conditioning remains broadly flat and poor across the practical eccentricity range.

Two clarifications are essential. First, the finite floor in Conjecture 5 applies to the *oblate curvature-resolving* model, not to the equivalent-sphere reduction of Conjecture 2; the latter remains exactly rank deficient in the analytic limit. Second, the observed effective power-law fits over moderate eccentricity windows are descriptive rather than asymptotic. They summarise a finite interval; they do not reveal a true singular law at the sphere. The mathematics here is gentler than the log–log plot first suggests.

3.5 Forward adequacy and inverse adequacy are different questions

Reduced models are often judged by forward error alone. Let $\mathbf{f}_{\text{red}}$ denote a reduced model and $\mathbf{f}_{\text{full}}$ a geometry-resolving model. A natural forward metric is the relative frequency misfit

$$\mathcal{E}_{\text{fwd}} = \frac{\|\mathbf{f}_{\text{red}}(\boldsymbol{\theta}) - \mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}{\|\mathbf{f}_{\text{full}}(\boldsymbol{\theta})\|_2}. \tag{21}$$

An inverse metric, by contrast, must assess the conditioning of the parameter recovery map; locally, $\kappa(J_s)$ is the appropriate quantity. There is no general theorem forcing small $\mathcal{E}_{\text{fwd}}$ to imply moderate κ. Indeed, the Browntone sequence provides counterexamples in both directions.

Conjecture 6 (Forward adequacy does not imply inverse adequacy) *There exist reduced models for fluid-filled viscoelastic shells such that*

$$\mathcal{E}_{\text{fwd}} \ll 1 \tag{22}$$

over an experimentally relevant parameter range, while the corresponding inverse problem for $\boldsymbol{\theta} = (a, c, E)$ is unusable because

$$\kappa(J_s) \gg 1 \qquad \textit{or} \qquad \operatorname{rank}(J_s) < 3. \tag{23}$$

Equivalently, a model may predict frequencies accurately enough for screening, classification, or single-parameter calibration, yet fail catastrophically as a tool for simultaneous recovery of geometry and material properties.

Paper 7 is the benign case. There the equivalent-sphere reduction is adequate because the inverse target is essentially one-dimensional: the geometry is measured independently, the dominant mode is identified, and the task is to map that frequency to an effective rind modulus [13]. Forward adequacy is enough because the geometric uncertainty has already been removed by other means. Paper 8 is the hostile case. There the objective is to recover (a, c, E) jointly from several modal frequencies, and the equivalent-sphere model fails precisely because its forward map suppresses one geometric direction [14]. The same simplification that is serviceable in Paper 7 is therefore defective in Paper 8. This is not a contradiction. It is the point.

Placeholder for the cross-application comparison figure.
Planned content: abdomen, bladder, watermelon, and borborygmi arranged on a common geometry–frequency–inference map.

Figure 1: Placeholder for the cross-application synthesis figure. The final version will place the four applications on a common geometry–spectrum– inference diagram.

The distinction also complements Paper 1. There the central lesson was that a correct resonance frequency does not by itself determine the dominant excitation pathway, because geometry also filters the forcing field [12]. Here the analogous lesson is that a correct frequency prediction does not by itself determine inverse usability, because geometry also filters the parameter directions visible to the data. In both cases, geometry mediates more than the location of the peaks.

Taken together, Conjectures 2– 6 define the capstone thesis in mathematical form. Scalar geometric reductions collapse one inverse direction; oblate curvature can restore it; the restoration is regular rather than singular near the sphere; and forward success is not a licence for inverse use. The results section tests these statements against the cross-paper evidence.

4 Results

We now test the four conjectures against the cross-application record assembled in Papers 1, 4, 5, 7, and 8. The point of the comparison is not that every application solves the same inverse problem. Quite the opposite. Some cases are dominated by excitation filtering, some by spectral organisation, and some by parameter recovery. What persists across them is that geometry decides which of those questions can be asked sensibly in the first place. Figure 1 is intended to visualise this comparison, while Table 1 collects the principal application-level evidence used below.

Table 1 already hints at the main result. The abdominal, bladder, watermelon, and borborygmi studies occupy different frequency bands and involve different observables, yet they all distinguish between a forward spectral pattern and the stronger claim that parameters can be recovered independently from that pattern. That distinction becomes explicit once the four conjectures are read as a sequence rather than as isolated claims.

4.1 Conjecture 1: equivalent-radius reductions are rank deficient

The first conjecture is the cleanest mathematically and, for that reason, the least forgiving. If the flexural spectrum depends on the geometry only through the scalar reduction

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{24}$$

then the two geometric sensitivity directions cannot remain independent. By the chain rule, every modal derivative with respect to the equatorial semi-axis a is proportional to the corresponding derivative with respect to the polar semi-axis c. In the scaled Jacobian J_s, the two geometry columns are therefore collinear. At the sphere this becomes physically transparent: a small perturbation in a and a small perturbation in c both act only by changing the same radius, so the geometry columns become proportional and one geometric direction disappears from the inverse map.

Paper 8 made that loss concrete for the canonical abdominal parameter set. The reported condition number for the equivalent-sphere inverse problem is

$$\kappa_{\text{sphere}} \approx 1.37 \times 10^{10}, \tag{25}$$

numerically finite only because the derivatives were estimated with a finite difference rather than evaluated symbolically [14]. The true statement is stronger than "poorly conditioned": the model architecture has already removed one independent geometric degree of freedom before any data are fitted. Figure 1 is intended to make this point visually by placing the equivalent-sphere reductions beside the curvature-aware models that follow.

The cross-application evidence supports this interpretation rather than weakens it. In Paper 7 the equivalent-sphere watermelon model works because the inverse task is deliberately narrow: the fruit geometry is measured separately, and the frequency is used to infer one effective stiffness scale E_{rind} [13]. In Paper 5 the constrained-bubble borborygmi model is similarly successful only because the dominant unknown is effectively the gas-pocket size or mode family, not the simultaneous recovery of two shape axes and a wall modulus [11]. These are not counterexamples to rank deficiency. They are examples of problems that avoid it by reducing the inverse target. Once one asks for joint recovery of (a, c, E) from spectra alone, the equivalent-radius reduction has already spent too much of the available information.

4.2 Conjecture 2: oblate asphericity lifts identifiability

The second conjecture asks whether this degeneracy is merely diagnosed or actually repairable. The evidence says that it is repairable, but only when the model retains genuine curvature information. In the oblate Ritz family of Paper 8, we may sweep continuously from the spherical limit toward the canonical abdominal geometry by decreasing the aspect ratio $\zeta = c/a$ from near unity to $\zeta = 0.667$, equivalently increasing the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$ from almost zero to the canonical oblate value $\varepsilon = 0.745$ [14]. Figure 2 is intended to show this sweep directly.

Two comparisons matter. The first is the broad conceptual one: the condition number drops from the 1×10^{10}-scale equivalent-sphere pathology to

$$\kappa_{\text{oblate}} = 69.4 \tag{26}$$

once the geometry is treated as oblate rather than spherical. The second is the stricter like-for-like comparison within the same oblate Ritz model: the conditioning improves from the near-spherical floor to the canonical oblate point without changing the inverse variables or the measured mode set. In both comparisons the message is the same. Asphericity is not merely decorative; it reintroduces a second geometric sensitivity channel.

The physical reason is mode-dependent curvature sampling. The lowest abdominal flexural mode already sits at

$$f_2 = 3.95\,\text{Hz}, \tag{27}$$

so the improvement in identifiability does not arise because the resonance is dragged into a more convenient frequency band. It arises because the higher flexural modes do *not* sample the oblate shell in the same way. Meridional and circumferential curvature changes are weighted differently by different mode shapes, so the a- and c-columns of J_s cease to be parallel. The bladder

Placeholder for the conditioning sweep figure.
Planned content: κ versus aspect ratio or eccentricity, showing the descent from the spherical limit to the canonical oblate shell.

Figure 2: Placeholder for the conditioning sweep from the near-spherical limit to the canonical oblate geometry. The final figure will show how retaining curvature information lifts the conditioning from the equivalent-sphere pathology to $\kappa_{\text{oblate}} = 69.4$.

study provides a useful supporting contrast here. Its fill-dependent U-shaped curve shows that geometry and stiffness can compete strongly in the forward spectrum [15]; Paper 8 shows what is additionally needed for inversion, namely that different modes must separate those effects rather than merely combine them.

This is also where the abdominal coupling result matters conceptually. In Paper 1 the same oblate geometry that yields the canonical $f_2 = 3.95\,\text{Hz}$ also produces a severe excitation asymmetry, with a reported mechanical-to-airborne displacement ratio $R \approx 3.3 \times 10^4$ in the canonical comparison [12]. The same geometry therefore controls both whether a mode is efficiently excited and whether it is diagnostically informative. That is the capstone thesis in one line: geometry decides what the system feels and what the experimenter can learn.

4.3 Conjecture 3: near-spherical asymptotics classify the inverse problem

The third conjecture concerns the approach to the sphere. A tempting but incorrect intuition is that the oblate inverse problem should obey a singular power law all the way to $\varepsilon = 0$. The Paper 8 computations show something milder and more informative. In the oblate Ritz family, the smallest singular value follows the regular even expansion

$$\sigma_{\min}(\varepsilon) = \sigma_0 + \lambda_1 \varepsilon^2 + O(\varepsilon^4), \tag{28}$$

with a positive intercept $\sigma_0 > 0$ over the fitted near-spherical regime [14]. The immediate consequence is that the oblate branch has a finite conditioning floor rather than a genuine asymptotic blow-up. Direct evaluation at $\varepsilon = 0.01$ gives

$$\kappa_{\text{floor}} \approx 269. \tag{29}$$

This is still awkward enough to make an experimentalist sigh, but it is not the same thing as algebraic singularity.

The resulting classification has three numerically distinct classes, as sketched in Figure 3. The first class is the *singular sphere*: the equivalent-sphere reduction has $\kappa_{\text{sphere}} \approx 1.37 \times 10^{10}$ because the inverse map is rank deficient. The second is the *regular bounded oblate* class: the curvature-resolving oblate family remains finite near the sphere, with $\kappa_{\text{floor}} \approx 269$, and improves to 69.4 at the canonical oblate point. The third is the *non-lifting prolate* class: breaking spherical symmetry in the wrong geometric direction does not help much, and the condition number stays in the range

$$\kappa_{\text{prolate}} \approx 561\text{–}729 \tag{30}$$

Placeholder for the near-spherical taxonomy figure.
Planned content: singular sphere, regular bounded oblate branch, and non-lifting prolate branch shown on the same conditioning plot.

Figure 3: Placeholder for the near-spherical conditioning taxonomy. The final figure will distinguish the singular equivalent sphere, the regular bounded oblate branch, and the non-lifting prolate branch.

across the tested prolate branch [14, 17].

This three-class picture matters because it replaces a vague slogan — "non-spherical is better" — with a more precise statement. Some departures from the sphere restore useful inverse information; some do not; and some model reductions remain singular regardless of numerical precision. The borborygmi and bladder papers again provide supporting context. Both show that spectra in soft, fluid-loaded biological systems can change strongly with geometry, fill state, or confinement [15, 11]. The near-spherical analysis tells us when such spectral variation should also be expected to carry independent inverse information. Those are related questions, but they are not the same question.

4.4 Conjecture 4: forward adequacy does not imply inverse adequacy

The fourth conjecture is the most practical, and Paper 7 supplies the most useful cautionary example. The equivalent-sphere watermelon model predicts tap-tone frequencies quite reasonably over the relevant stiffness range: f_2 spans 46.4 Hz to 203.2 Hz, and at the canonical reference point the model gives $f_2 = 89.9$ Hz with a clean monotonic relation between f_2^2 and the effective rind modulus E_{rind} [13]. For that task the reduced model is perfectly sensible. Geometry is measured independently; the inverse target is essentially one-dimensional; and the model therefore serves as a practical forward predictor and calibration law.

The trouble begins only when that forward success is promoted into a general inversion claim. If the same equivalent-sphere architecture is asked to recover (a, c, E) jointly from modal data, the condition number returns to the 1×10^{10} scale, because the geometry has again been collapsed into a single radius. The curvature-resolving oblate model, by contrast, yields broadly similar low-order frequencies while restoring a usable inverse map with

$$\kappa = 69.4. \tag{31}$$

Figure 4 is intended to show this distinction: small forward spectral error and good inverse conditioning are related only loosely, and sometimes not at all.

The other applications reinforce the same lesson from different angles. The bladder model predicts a clear forward phenomenon — a minimum $f_2 = 13.5$ Hz at 222 mL — and a clear coupling phenomenon — $\mathcal{R}_{\text{bladder}} \approx 6{,}400$ in favour of mechanical excitation [15]. Neither result, by itself, certifies that a joint inverse problem for geometry and material parameters would be well conditioned. The borborygmi model similarly supports narrow inferences, such as mapping a peak near 340 Hz to a pocket volume of order 3 mL, without claiming that bowel geometry and wall elasticity can be reconstructed uniquely from the sound alone [11]. Forward adequacy is often enough for diagnosis, classification, or design. It is not a licence to ignore identifiability.

Placeholder for the forward-versus-inverse adequacy figure.
Planned content: comparable low-order frequency predictions for reduced and oblate models alongside sharply different condition numbers.

Figure 4: Placeholder for the distinction between forward adequacy and inverse adequacy. The final figure will compare frequency prediction error against inverse conditioning for the reduced and curvature-aware models.

Taken together, the four conjectures survive the cross-application synthesis. Equivalent-radius reductions are structurally rank deficient; oblate curvature can lift that defect; the lifting is regular rather than singular near the sphere; and forward predictive success does not guarantee inverse usefulness. That is why the same research programme can contain, without contradiction, a successful equivalent-sphere watermelon model and a strong warning against equivalent-sphere inversion. The models answer different questions. Geometry, as ever, decides which answers are trustworthy.

5 Discussion

6 Conclusion

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Table 1: Cross-application evidence for the geometry–spectrum–inference thesis. Here f_2 denotes the lowest flexural mode, R_{eq} the equivalent radius where used, and κ the condition number of the scaled Jacobian for the local inverse problem.

Application	Geometry and key parameters	Characteristic frequencies	Coupling / inference takeaway
Abdomen	Canonical oblate shell: $a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 0.01\,\mathrm{m}$, $E = 0.1\,\mathrm{MPa}$, $\nu = 0.45$, $\rho_f = 1020\,\mathrm{kg/m^3}$, $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$	$f_2 = 3.95\,\mathrm{Hz}$; higher flexural modes in the 4 Hz to 10 Hz band; breathing mode $n = 0$ near 2490 Hz	Geometry filters forcing as well as spectra: the reported mechanical-to-airborne displacement ratio is $R \approx 3.3 \times 10^4$ in the canonical comparison; the same geometry underpins the identifiability study with $\kappa_{\mathrm{oblate}} = 69.4$
Bladder	Fill-dependent near-spherical shell over 50 mL to 500 mL; representative 300 mL state: $R = 4.15\,\mathrm{cm}$, $h = 1.79\,\mathrm{mm}$, $E = 70.9\,\mathrm{kPa}$	U-shaped $f_2(V)$ curve with a minimum $f_2 = 13.5\,\mathrm{Hz}$ at 222 mL; representative 300 mL value $f_2 = 13.9\,\mathrm{Hz}$	Geometry and strain-stiffening jointly organise the spectrum; mechanical coupling exceeds airborne coupling by $\mathcal{R}_{\mathrm{bladder}} \approx 6{,}400$; useful fill-state inference is possible, but no full geometry–material inversion is claimed
Watermelon	Oblate fruit approximated by an equivalent sphere: $a = 0.158\,\mathrm{m}$, $c = 0.123\,\mathrm{m}$, $h = 15\,\mathrm{mm}$, $R_{\mathrm{eq}} = 0.1453\,\mathrm{m}$, $E_{\mathrm{rind}} = 10\,\mathrm{MPa}$ to $250\,\mathrm{MPa}$	Tap-tone flexural frequency $f_2 = 46.4\,\mathrm{Hz}$ to $203.2\,\mathrm{Hz}$ across the constructed stiffness sweep; canonical example $f_2 = 89.9\,\mathrm{Hz}$ at $E_{\mathrm{rind}} = 50\,\mathrm{MPa}$	Equivalent-sphere reduction is forward-useful when geometry is measured independently and the inverse target is one effective stiffness parameter; that same reduction is unsuitable for joint recovery of (a, c, E)
Borborygmi	Quasi-spherical or slug-like gas pockets in a 30 mm lumen; canonical constrained-bubble parameters: $V = 1\,\mathrm{mL}$ to $50\,\mathrm{mL}$, $E = 10\,\mathrm{kPa}$, $h = 3\,\mathrm{mm}$	Constrained-bubble resonance 135 Hz to 440 Hz; canonical 10 mL pocket: 223 Hz; healthy spectral peak near 340 Hz implies $V \sim 3\,\mathrm{mL}$	Geometry selects the active resonance family; narrow inference about pocket size or mechanism is plausible, but only because the inverse task is low-dimensional rather than a full shell identification problem